

BUSINESS REQUIREMENTS DOCUMENT

Linux Cluster Monitoring System

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Revision History

Name	Date	Reason For Changes	Version
Edward Wang	January 1, 2022	Oversimplified initial draft for internal discussion purposes	v0.1
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1. Executive Summary

The organization is looking to acquire and set up a Linux Cluster Resource Monitoring System to track hardware use across its data center. This system aims to give the Linux Cluster Administration (LCA) team the data needed for effective server capacity planning. With this information, they can make informed decisions about adding or removing servers.

The project is set to finish by June 1, 2026. It will follow a phased approach that includes requirements gathering, development, testing across multiple environments, production deployment, and knowledge transfer. The monitoring system will be installed on all 500 Linux servers in the cluster. It will start providing daily hardware usage reports and automated alerts when it goes live.

The main requirements for this project cover four key areas: data collection (gathering hardware metrics from each node), data storage (a database to keep historical usage data), reporting and analytics (automated daily reports for the LCA team), and alerting (automated notifications when a server goes offline or exceeds set usage limits). A subscription management website will also be created to help stakeholders manage their report and alert preferences.

The stakeholders involved in this project include the Project Manager, the Linux Cluster Administration team, the business owner, the Implementation and Agile development team responsible for building and testing the system, and the Support team responsible for deployment and ongoing maintenance. This document outlines the project objectives, business needs, scope boundaries, functional requirements, stakeholder responsibilities, project schedule, and a cost-benefit analysis to back the investment decision.

2. Project Objectives

The main goal of this project is to create a fully functional Linux Cluster Monitoring System. This system will help the LCA team make timely, data-driven decisions about managing hardware resources. The following objectives are defined using the SMART framework to make sure they are specific, measurable, achievable, relevant, and time-bound.

2.1 Specific

The project will provide a monitoring agent installed on all 500 Linux servers in the organization's data center. It will include a centralized database for storing collected hardware metrics, an analytics and reporting module that can generate daily usage summaries, an automated alert system for offline and threshold-breach events, and a self-service website where stakeholders can subscribe to or unsubscribe from reports and alerts.

2.2 Measurable

Success will be measured by the full installation of the monitoring agent on all 500 servers, the delivery of daily hardware reports before 6:10 PM each business day, and the reliable sending of threshold alerts within an acceptable response time when CPU or RAM usage on any server exceeds 90 percent or when a server is detected as offline.

2.3 Achievable

The project team includes a dedicated Agile team comprising one Scrum Master, one Product Owner, one Business Systems Analyst, two developers, and one QA specialist. The timeline includes development, testing across four environments, and a knowledge transfer period. This makes the June 1, 2026, target achievable if resources are properly allocated and stakeholders are available.

2.4 Relevant

Hardware capacity planning is an important task for the organization. Without automated monitoring, the LCA team must manually check server usage across 500 nodes. This process takes a lot of time, is error-prone, and does not support quick incident response. This project aims to fill that gap and supports the organization's larger goal of improving efficiency and lowering infrastructure risk.

2.5 Time-Bound

The project needs to be fully operational and provide reports and alerts by June 1, 2026. All intermediate milestones, including requirements sign-off, MVP delivery, implementation, testing, and deployment, must be finished before this date. You'll find the details in the project schedule section of this document.

3. Needs Statement

The Linux Cluster Administration team manages a cluster of 500 Linux servers in the organization's data center. The team's duties include planning server capacity, managing hardware lifecycles, and improving infrastructure. To do this well, the LCA team needs clear and timely information on each server's hardware usage in the cluster.

Currently, the lack of an automated monitoring solution forces the team to rely on manual methods to monitor server health and performance. This method is not workable across 500 nodes, delays decision-making, and increases the chance of service interruptions due to unnoticed hardware strain or server failures.

The LCA team needs daily hardware usage reports. These reports should cover key metrics such as CPU usage, RAM usage, disk input/output rates, and disk space availability. They will help in deciding when to add servers to the cluster if it is overloaded or remove servers when the cluster is underused, thus reducing infrastructure costs.

In addition to scheduled reports, the LCA team requires real-time alerts. The system must send automated email notifications when any server goes offline or when the CPU or RAM usage on any server exceeds 90 percent. These alerts will help the team respond quickly to performance issues or hardware failures before they lead to service outages.

Implementing an automated Linux Cluster Monitoring System is not just a luxury; it is essential for the business. Without it, the organization risks spending too much on unnecessary hardware, not having enough hardware, leading to system instability, and reacting to failures instead of preventing them.

4. Project Scope

4.1 In Scope

The following work is part of this project. The team will evaluate and select the right technologies to collect hardware performance data from each of the 500 nodes in the Linux cluster. They will store the data in a permanent database and analyze it for reporting. A monitoring agent will be developed and installed on each server to gather the necessary metrics at set intervals.

The project also includes creating a web-based subscription portal. This portal will allow stakeholders to subscribe to or unsubscribe from daily reports and real-time alerts. The system will be tested in four deployment environments: DEV (development), QC/TEST (quality control and testing), PRE-PROD (pre-production testing), and PROD (production). This will follow the organization's standard deployment process.

Once the implementation is complete, the Support team will receive a formal knowledge-transfer session covering system design, agent installation steps, configuration management, and maintenance duties. The LCA team will get supporting documents and operational manuals to help them use the system moving forward.

4.2 Out of Scope

The following items are clearly excluded from this project and will not be addressed unless a formal change request is approved. Application-level performance monitoring, including tracking the performance of individual software applications running on servers, is out of scope. The system will monitor only hardware-level metrics. Network traffic analysis and intrusion detection are also excluded because these tasks fall within the responsibilities of the organization's network and security teams.

Monitoring of cloud-hosted or third-party managed servers is not included. This system is designed solely for the on-premises Linux cluster. Custom hardware diagnostic tools, tracking for physical server repairs, and any automated fixes or auto-scaling features are not part of this initial phase. Integration with external IT service management (ITSM) platforms such as ServiceNow or Jira is also outside the current project scope, though it may be considered for a future phase.

5. Requirements

5.1 Functional Requirements

The following functional requirements define the capabilities the system must provide to meet the business needs of the Linux Cluster Administration team.

5.1.1 Data Collection

The system must include a lightweight agent program installed on each of the 500 Linux servers in the cluster. This agent is responsible for regularly collecting hardware performance data from its host server. The metrics to be collected include, at minimum, CPU utilization percentage, RAM usage and availability, disk input/output rates, disk space consumption, and network interface status. The agent must operate in the background with minimal impact on server performance and must be compatible with the Linux distributions used in the cluster. Possible technologies for the agent include Bash scripts, Python, or Java. The final choice will be made during the technical design phase based on performance, maintainability, and compatibility.

5.1.2 Data Storage

Collected hardware metrics must be stored in a centralized database that both the analytics and reporting module and the administrative team can access. The database must handle the data generated by 500 servers that report daily. It must also keep historical data for trend analysis and capacity planning over time. The right database technology, whether a relational database management system (RDBMS) or a NoSQL solution, will be chosen during the design phase. This choice will depend on scalability, query performance, and operational needs.

5.1.3 Reporting and Analytics

The system must create a daily hardware usage report for all servers in the cluster. This report needs to be sent to LCA team members who subscribed by 6:10 PM each day. It must show hardware use data in a clear, easy-to-read format and point out any servers that are nearing or exceeding set usage limits. The analytics component should help analyze historical trends to guide capacity planning decisions. Possible technologies include SQL queries, Excel, Tableau, or Python data processing pipelines.

5.1.4 Alerting

The system must have a real-time alerting mechanism that sends email notifications to subscribed stakeholders under two conditions. First, an alert must be sent when any server in the cluster goes offline, which may signal a hardware failure or connectivity issue. Second, an alert must be sent when CPU or RAM usage on any server exceeds 90 percent, indicating a risk of performance degradation. Alert delivery times and escalation procedures will be finalized in the Software Requirements Document with help from the LCA team.

5.1.5 Subscription Management Portal

A web-based portal must be created to allow users to manage their subscriptions to daily reports and real-time alerts. The portal must let individual users subscribe and unsubscribe from report distribution lists and alert notifications independently. Access to the portal must be limited to authorized personnel, and user subscription preferences must be stored in the system.

database. The portal must be reachable from standard web browsers without needing any extra software installation.

5.2 Non-Functional Requirements

The following non-functional requirements define the quality attributes, performance characteristics, and operational constraints the system must satisfy. These requirements apply across all functional areas described above.

5.2.1 Performance and Timeliness

The monitoring agent must operate in the background with minimal CPU and memory footprint so that it does not degrade the performance of the host server. The daily hardware usage report must be generated and delivered to subscribed LCA team members by 6:10 PM each business day. Threshold and offline alerts must be delivered promptly after the triggering condition is detected so the team can respond before a performance issue escalates.

5.2.2 Availability and Reliability

The monitoring system must be reliable enough that scheduled reports and real-time alerts are not missed. The collection agent must resume metric collection automatically following a host server reboot or transient failure, and the central collection and storage services must remain available throughout standard business operations.

5.2.3 Scalability

The system must handle the volume of metrics generated by all 500 nodes reporting on a recurring schedule, and must accommodate growth of the cluster as servers are added without requiring a redesign. The central database must retain historical usage data to support long-term trend analysis and capacity planning.

5.2.4 Security and Access Control

Access to the subscription management portal must be limited to authorized personnel. User subscription preferences and collected hardware metrics must be stored and transmitted in accordance with the organization standard security practices for internal infrastructure systems.

5.2.5 Compatibility and Maintainability

The monitoring agent must be compatible with the Linux distributions deployed across the cluster. The subscription portal must be reachable from standard web browsers without requiring any additional client software. The system must be documented sufficiently to support a formal knowledge transfer to the Support team for ongoing maintenance.

6. Key Stakeholders

The following stakeholders have been identified as key participants in the planning, development, deployment, and ongoing operation of the Linux Cluster Monitoring System. Each stakeholder group carries distinct responsibilities and interests in the outcome of this project.

Stakeholder	Role	Responsibilities	Interest Level
Project Manager	Oversight	Holds all parties accountable to the project timeline and budget	High
Linux Cluster Administration (LCA) Team	Business Owner	Provides business requirements; consumes reports and alerts; funds implementation and maintenance	High
Implementation / Agile Team	Development	Designs, develops, and tests the monitoring system across all environments	High
Support Team	Operations	Deploys and maintains the system in production; provides Level 2 support	Medium

The Project Manager coordinates all stakeholder groups, manages the project timeline, and ensures deliverables are completed on schedule and within the defined scope. The Project Manager is the main point for escalating cross-team dependencies and risks.

The Linux Cluster Administration (LCA) team acts as the business owner of this project. The LCA team defines the business requirements, including the specific hardware metrics to monitor, the report format and delivery schedule, and the alert threshold values. This team will also be the main user of the system's outputs and will cover implementation and ongoing maintenance costs.

The Implementation and Agile team is responsible for turning the business requirements into a working technical solution. The team includes one Scrum Master, one Product Owner (who may also be the Project Manager), one Business Systems Analyst, two software developers, and one QA specialist. This team follows Agile development practices, providing incremental functionality throughout the project lifecycle.

The Support team handles the project's operational aspects, including deploying the monitoring agent to all 500 servers, setting up the production infrastructure, and taking responsibility for the system after launch. Ongoing tasks include applying software upgrades, monitoring the health of the monitoring system, and offering Level 2 technical support for issues that go beyond the LCA team's capabilities.

7. Schedule, Timeline, and Milestones

The project is structured into seven sequential milestones, each with a defined owner and target completion date. The overall project must be fully operational by June 1, 2026. The milestones are designed to account for dependencies between phases; for example, implementation cannot begin until requirements are finalized, and production deployment cannot proceed until testing across all prior environments is successfully completed. A buffer of approximately two weeks has been built into the timeline between the completion of implementation and the production deployment date to accommodate unforeseen issues during testing.

#	Target Date	Milestone	Owner
1	March 1, 2026	Gather and finalize business requirements from all stakeholders	BSA / Project Manager
2	March 21, 2026	Deliver a Minimum Viable Product (MVP) for stakeholder review	Implementation / Agile Team
3	May 1, 2026	Complete full implementation of the monitoring system	Implementation / Agile Team
4	May 15, 2026	Complete testing across DEV, QC/TEST, and PRE-PROD (PAT) environments	QA - Agile Team
5	May 21, 2026	Deploy the system to the Production environment	Support Team
6	May 31, 2026	Complete knowledge transfer and training with the LCA and Support teams	Implementation Team
7	June 1, 2026	Deliver the first cluster hardware usage reports and threshold alerts to the LCA team	Support Team

It is important to note that stakeholder availability for requirements gathering, and the knowledge transfer session poses a significant scheduling challenge. Any delay in securing LCA team participation during the requirements phase could affect the MVP and implementation milestones. The Project Manager is responsible for tracking these issues and informing the project sponsor if the timeline is at risk.

8. Cost-Benefit Analysis

8.1 Costs

The implementation of the Linux Cluster Monitoring System will incur both one-time and recurring costs. One-time costs include the labor required from the Implementation and Agile team, encompassing the time of two developers, one QA specialist, one BSA, one Scrum Master, and one Product Owner over the project duration of approximately three months. These costs also include the infrastructure required to host the monitoring database and web portal in the production environment.

Recurring costs following go-live include the labor cost of the Support team for ongoing system maintenance, including software upgrades, health monitoring, and Level 2 support. Any licensing fees associated with the selected analytics or database technology will also contribute to the ongoing cost base. These recurring costs, however, are expected to be substantially lower than the costs of the current manual monitoring approach.

8.2 Benefits

The primary financial benefit of this system is the elimination of manual hardware monitoring across 500 servers. The time currently expended by the LCA team in manually collecting and reviewing server performance data represents a significant and recurring operational cost. Automating this process will redirect that effort toward higher-value activities such as strategic capacity planning and infrastructure optimization.

A second material benefit is the optimization of hardware procurement and decommissioning decisions. By providing accurate and timely utilization data, the system will reduce the risk of over-provisioning, where unnecessary servers are purchased at significant capital expense, and under-provisioning, where insufficient capacity leads to performance degradation and service interruptions. Data-driven provisioning decisions are expected to yield measurable reductions in hardware expenditure over time.

The real-time alerting capability provides a third category of benefit by enabling proactive incident response. Early detection of server failures or performance degradation reduces mean time to resolution (MTTR) and minimizes the risk of prolonged service outages. The business impact of unplanned downtime, including lost productivity, potential SLA penalties, and reputational risk, is difficult to quantify precisely but is understood to be material. The alerting system directly mitigates this exposure.

In aggregate, the expected benefits of reduced manual labor, optimized hardware investment, and improved operational resilience are assessed to substantially outweigh the implementation and maintenance costs of the proposed system, supporting a favorable return on investment over a multi-year horizon.

9. Document Approval

This Business Requirements Document has been prepared by the Business Systems Analyst and is submitted for review and approval by the designated stakeholders listed below. By signing this document, approvers confirm that the requirements, scope, schedule, and cost-benefit analysis accurately reflect the agreed-upon business needs and project parameters.

Role	Name	Date
Project Manager		
LCA Team Lead		
Implementation Team Lead		